

MODEL RESEARCH DOCUMENT

Project Title -Powered Workforce Analytics & Talent Intelligence Dashboard

Introduction

This document lists machine learning, deep learning, ensemble, clustering, anomaly detection, and generative AI models that can strengthen predictive workforce analytics, employee performance analysis, talent intelligence, skill-gap detection, segmentation, anomaly detection, and AI-assisted reporting for structured enterprise HR datasets

Models and Technologies

Model / Technology	Use Case	Advantages	Expected Performance
Extra Trees	Employee risk prediction	Fast ensemble, low variance, robust to noise	92–96% Accuracy
XGBoost	Attrition prediction, employee classification	High accuracy, handles missing values, excellent for structured data	94–97% Accuracy
LightGBM	Large-scale workforce analytics	Fast training, memory efficient, excellent performance	94–97% Accuracy
CatBoost	HR datasets with categorical features	Handles categorical variables with minimal preprocessing	94–98% Accuracy
TabNet	Workforce analytics on structured data	Interpretable deep learning model for tabular datasets	94–97% Accuracy
FT-Transformer	Structured workforce analytics	Learns complex feature interactions with high accuracy	95–98% Accuracy
TabTransformer	Mixed numerical and categorical HR data	Context-aware feature embeddings	94–98% Accuracy
Voting Classifier	Ensemble workforce prediction	Combines multiple models for stable predictions	93–97% Accuracy
Stacking Classifier	Advanced ensemble learning	Improves generalization using meta-learning	95–98% Accuracy
Isolation Forest	Workforce anomaly detection	Detects outliers without labeled data	High Precision
Local Outlier Factor (LOF)	Employee behavior analysis	Density-based anomaly detection	High Precision
Gaussian Mixture Model (GMM)	Employee segmentation	Probabilistic clustering for flexible grouping	Good Clustering Performance

UMAP	Data visualization and dimensionality reduction	Preserves local structure while reducing dimensions	Excellent Visualization
Autoencoder	Feature learning and anomaly detection	Learns hidden representations for anomaly scoring	High Performance
RoBERTa	Resume parsing and employee feedback analysis	Strong contextual NLP understanding	F1 Score: 92–95%
DeBERTa	Skill extraction and HR document analysis	Improved contextual embeddings	F1 Score: 93–96%
FLAN-T5	HR question answering	Instruction-tuned model for enterprise Q&A	Benchmark-Level Performance
Gemma	Enterprise reporting assistant	Lightweight and efficient LLM	Benchmark-Level Performance
Phi-3	Lightweight HR assistant	Efficient reasoning with low computational requirements	Benchmark-Level Performance
Qwen	Multilingual workforce analytics	Strong multilingual reasoning capabilities	Benchmark-Level Performance
DeepSeek	Analytics, reasoning, and code generation	Excellent analytical and reasoning performance	Benchmark-Level Performance
Retrieval-Augmented Generation (RAG)	Knowledge-grounded workforce question answering	Retrieves enterprise HR documents to generate accurate, explainable responses	Highly Recommended for Enterprise AI

Conclusion

The proposed models and technologies significantly strengthen the AI-Powered Workforce Analytics & Talent Intelligence Dashboard by improving prediction, segmentation, anomaly detection, and intelligent reporting. Together, they enable more accurate workforce insights, smarter HR decision-making, and a more proactive, data-driven approach to talent management.